Macquarie Master Plan
Bushfire

Prepared for
Department of Planning, Housing and Infrastructure

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Draft Status

This document has been prepared by Blackash Bushfire Consulting as a working draft to support the ongoing development of the Macquarie Mine Master Plan and associated Enquiry by Design (EbD) process. The document is intended to facilitate discussion, testing of planning assumptions and integration of bushfire considerations into the broader strategic planning framework. The contents of this draft, including any opinions, conclusions, findings, recommendations or planning directions, are preliminary in nature and should not be relied upon for any purpose other than project review and stakeholder consultation. The document remains subject to further investigation, refinement, review of supporting technical studies, agency consultation and consideration of future planning outcomes.Blackash Bushfire Consulting reserves the right to amend, revise, supplement or withdraw any part of this document as additional information becomes available or project requirements evolve. To the fullest extent permitted by law, Blackash Bushfire Consulting accepts no responsibility or liability arising from reliance upon this draft document or any decisions made on the basis of information contained herein prior to finalisation of the report.This document should be read in conjunction with the broader suite of technical studies, specialist investigations and planning documents prepared to support the Macquarie Mine Master Plan process.

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Contents

1. Project Overview

1.1 Purpose

This bushfire technical assessment has been prepared to support the post-mining land use planning and master planning process for the Macquarie Mine precinct. The assessment forms part of the broader technical workstream being undertaken to inform the development of the Draft Structure Plan and establish a strategic planning framework for future land use opportunities across the site.

Consistent with the Post Mining Land Use (PMLU) program, the assessment is deliberately scoped at a rezoning and master planning level rather than a development application level. The purpose of the assessment is not to undertake detailed bushfire design, Bushfire Attack Level (BAL) assessments, emergency management planning or development-specific compliance analysis. Rather, the assessment seeks to identify the key bushfire opportunities, constraints and resilience considerations that may influence future land use allocation, development capacity, staging and infrastructure planning.

The assessment adopts a constraints-led and risk-based planning approach, recognising that bushfire is one of several factors that will influence the long-term suitability of land for future development. Consistent with the Enquiry by Design (EbD) process, bushfire considerations are assessed alongside environmental constraints, rehabilitation objectives, biodiversity outcomes, infrastructure requirements, transport networks, servicing opportunities, cultural heritage considerations and broader strategic planning objectives.

1.2 Assessment Principles

The assessment has been guided by the core principles established for the PMLU program, with a focus on providing sufficient information to support strategic planning and master plan decision-making while avoiding unnecessary duplication of future development assessment processes.

The assessment therefore seeks to:

- identify bushfire constraints that may influence the sequencing, configuration and intensity of future development;
- leverage existing information including mine closure planning, rehabilitation strategies, previous technical investigations, bushfire mapping, fire history and outcomes from the EbD process;
- undertake targeted verification of key bushfire considerations where necessary to inform strategic planning decisions;
- assess opportunities and constraints in a practical manner, recognising that future development may be appropriate where bushfire risk can be effectively managed through land use planning and infrastructure provision;
- integrate bushfire considerations with other technical disciplines through precinct-scale analysis and spatial planning; and
- establish clear separation between strategic planning requirements and matters that should be deferred to future development application, subdivision and detailed design stages.

1.3 Role of the Bushfire Assessment

The assessment is intended to provide the bushfire evidence base necessary to support the Draft Structure Plan and future rezoning considerations. Specifically, the assessment seeks to:

- understand the broader bushfire landscape affecting the Macquarie precinct;
- identify areas where bushfire hazard may influence future land use allocation and development potential;

- assess the strategic feasibility of accommodating Asset Protection Zones (APZs), access arrangements and firefighting infrastructure;
- identify bushfire resilience measures that should be incorporated into the Structure Plan;
- inform development staging and infrastructure sequencing;
- establish bushfire-related objectives and performance criteria suitable for inclusion within the Master Plan; and
- provide a framework for future engagement with the NSW Rural Fire Service and other relevant agencies.

This report has been completed by Lew Short who is the Director of Blackash Bushfire Consulting Pty Ltd and an accredited BPAD Level 3 Practitioner (BPAD-A) (Accreditation No. BPD-PA-16373). He has over 25 years' experience in bushfire planning and risk assessment, providing strategic advice for planning proposals, State Significant Development, major subdivisions, renewable energy projects and complex performance based bushfire assessments across New South Wales.

1.4 Integrated Planning Approach

The bushfire assessment forms one component of a multidisciplinary planning process involving urban design, engineering, environmental, heritage, economic and statutory planning specialists. Through the EbD process, bushfire considerations are integrated with flooding, biodiversity, rehabilitation, transport, infrastructure, contamination, servicing and land use planning considerations to ensure that future development opportunities are assessed holistically rather than in isolation.

This integrated approach recognises that the future success of the Macquarie precinct will depend on balancing environmental, social, economic and resilience outcomes. Bushfire therefore represents both a constraint and a design consideration that must be addressed through the structure and layout of future development, the management of rehabilitation landscapes, the provision of access and emergency infrastructure, and the long-term stewardship of the site.

The outcome sought is a Structure Plan that is resilient, adaptable and capable of accommodating future land uses while responding appropriately to bushfire risk, environmental values, rehabilitation objectives and the broader strategic planning framework established under the PMLU program.

1.5 Methodology

The bushfire assessment was undertaken using a strategic, risk-based methodology consistent with the objectives of the EbD process and the requirements of PBP relating to strategic planning. The assessment commenced with a desktop review and strategic hazard mapping exercise to establish the bushfire context affecting the precinct and inform the development of the Draft Structure Plan. This included a review of bushfire prone land mapping, vegetation characteristics, topography, fire history, rehabilitation areas and the broader landscape setting to understand the distribution and influence of bushfire hazard across the site.

Strategic hazard screening was undertaken to identify potential fire run pathways, high-risk development interfaces and the implications of bushfire risk for future land use allocation, development staging and infrastructure planning. The assessment was undertaken in collaboration with ecology, rehabilitation, open space and infrastructure specialists to ensure that vegetation retention, environmental outcomes, rehabilitation objectives and bushfire protection requirements were considered in an integrated manner.

Building upon the strategic hazard assessment, a high-level review of APZ feasibility, emergency access and egress requirements, and firefighting infrastructure considerations was undertaken to support the preparation of the Draft Structure Plan. A key objective of this assessment was to determine whether future precincts could be activated and developed having regard to bushfire protection requirements and whether sufficient land was available to support the intended land uses, development capacity and long-term operation of each precinct.

The APZ assessment was undertaken at a master planning scale and focused on verifying that adequate land exists to accommodate future APZs, managed land, perimeter roads and associated bushfire protection measures consistent with the principles and intent of PBP. Rather than determining detailed APZ dimensions or development application stage compliance outcomes, the assessment was used to test precinct availability, identify any bushfire-related constraints to future activation, and inform land use allocation, staging and development capacity decisions.

Strategic access and egress requirements were similarly assessed throughout the EbD process to ensure that future precincts can be safely accessed, serviced and operated, while supporting emergency response, firefighting operations, evacuation and staged development. Collectively, this methodology enabled bushfire considerations to be integrated into the evolution of the Draft Structure Plan, ensuring that future precincts are capable of being activated and developed in a manner that is both practical and resilient, while providing a robust framework for long-term bushfire risk management at a precinct scale.

Vegetation and slope information affecting the precinct has been derived and assessed at a strategic level to understand the broad bushfire hazard context and inform the development of the Draft Structure Plan. This information has been used to identify potential hazard interfaces, understand likely bushfire exposure and verify that future development precincts are generally capable of accommodating bushfire protection measures consistent with bushfire requirements. However, detailed APZ assessments have not been undertaken as part of this study.

Given the strategic nature of the Draft Structure Plan and the absence of defined development footprints, building envelopes, subdivision layouts, final land uses and rehabilitation outcomes, it would be premature to determine detailed APZ widths or management requirements at this stage. Accordingly, the assessment has focused on APZ feasibility and precinct-scale land availability rather than detailed APZ design, with site-specific APZ calculations and compliance assessments to be undertaken during future precinct planning, subdivision and development application stages when sufficient design detail becomes available.

1.6 Constraints-Led Structure Planning Approach

The Draft Structure Plan has been developed using a constraints-led planning approach, recognising that the long-term success, resilience and deliverability of the precinct is dependent upon understanding and responding to site constraints before land use outcomes are determined. Rather than commencing with a development yield or land use outcome and subsequently attempting to resolve environmental, infrastructure and bushfire issues, the planning process has sought to identify the opportunities and limitations of the site at the outset and use these to inform the structure and configuration of future development.

This approach is consistent with contemporary strategic planning principles and the requirements of PBP, which requires bushfire risk to be considered at the earliest stages of planning and land use allocation. It also aligns with broader resilience-based planning principles that seek to avoid or minimise risk through appropriate site planning rather than relying solely on development-stage mitigation measures.

The Draft Structure Plan has been informed through an EbD process involving a multidisciplinary team of specialists. Through this collaborative process, each discipline has identified the constraints, opportunities and operational requirements relevant to their area of expertise, allowing the various functional elements of the precinct to be considered collectively rather than in isolation.

The planning process has therefore considered and integrated a range of factors including bushfire hazard, vegetation retention and rehabilitation, biodiversity values, topography, flooding and drainage, transport networks, utility infrastructure, servicing requirements, access and emergency management, operational and rehabilitation mining considerations, future land use compatibility and development staging. The interaction between these factors has informed the location, extent and function of future precincts and ensured that development areas are positioned within locations capable of accommodating the necessary infrastructure, environmental outcomes and risk management measures.

Importantly, the Structure Plan should not be viewed as a collection of independent land use precincts. Rather, it represents an integrated planning framework in which each functional area is connected through a coordinated network of roads, infrastructure corridors, environmental lands, rehabilitation areas, APZs, servicing networks and emergency management systems. This integration enables the precinct to function as a cohesive and resilient whole, while also providing flexibility for future detailed planning, subdivision and development delivery.

The outcome of this constraints-led approach is a Draft Structure Plan that seeks to balance development opportunities with environmental protection, operational requirements, infrastructure efficiency and community resilience. By identifying and responding to constraints early, the Structure Plan reduces approval risk, improves long-term deliverability and establishes a robust framework for future development that is capable of responding to changing operational, environmental and bushfire conditions over time.

2. Bushfire Planning Framework

2.1 Purpose and intended outcomes

This bushfire technical assessment has been prepared to support the rezoning and master planning process for the Macquarie Mine precinct (Figure 1). The assessment adopts a strategic, risk-based approach to understanding bushfire hazard and resilience and has been undertaken to inform land use allocation, development capacity, Asset Protection Zone (APZ) feasibility, access and egress requirements, emergency management considerations and future development staging.

Consistent with the outcomes of the Enquiry by Design (EbD) process, bushfire has been recognised as a primary design driver for the precinct, with extensive areas of the site and surrounding landscape affected by bush fire prone land and capable of influencing future development outcomes. The assessment therefore seeks to establish the bushfire planning framework necessary to guide the preparation of the Draft Structure Plan, translating the bushfire principles and requirements into strategic planning objectives, performance criteria and implementation measures.

Importantly, the assessment is intended to inform master planning and rezoning decisions and does not seek to duplicate the detailed Bushfire Attack Level (BAL), APZ design, emergency management or development-specific compliance assessments that will be required as part of future subdivision, development application and detailed design processes.

2.2 Bushfire Framework

This bushfire technical assessment has been prepared to inform the rezoning and master planning process for the precinct. The assessment adopts a strategic, risk-based approach to bushfire planning consistent with the requirements of *Planning for Bush Fire Protection 2019* (PBP), *Ministerial Direction 4.3 – Planning for Bushfire Protection*, and the principles of contemporary disaster resilience and land use planning.

The purpose of the assessment is not to undertake detailed development application level bushfire design, Bushfire Attack Level (BAL) assessment or building-specific compliance analysis. Rather, the assessment establishes the strategic bushfire planning framework necessary to inform land use allocation, development capacity, infrastructure planning and the long-term structure of the precinct.

Bushfire has been identified as a primary design driver for the precinct. Large portions of the site are mapped as bushfire prone land (Figure 2, 3 & 4) and contain existing and future vegetation that has the potential to influence development outcomes, infrastructure requirements and the overall capacity of the site to accommodate future land uses. The interaction between bushfire hazard, rehabilitation areas, retained vegetation, topography, access arrangements and proposed development creates a complex planning environment that must be addressed through the master planning process rather than deferred to future development stages.

The Draft Structure Plan has been informed by a series of Enquiry by Design (EbD) workshops involving planners, bushfire consultants, ecologists, engineers, infrastructure specialists, rehabilitation practitioners and relevant government agencies. The EbD process has facilitated the collaborative testing of land use configurations, development interfaces, access networks, rehabilitation outcomes and infrastructure requirements, enabling bushfire risk to be considered at the appropriate strategic planning scale. This process has ensured that bushfire protection requirements, including future Asset Protection Zones, access and evacuation arrangements, vegetation management considerations and land use compatibility, have been integrated into the development of the Draft Structure Plan and balanced alongside environmental, operational and development objectives.

Consistent with the strategic study requirements contained within PBP, this assessment seeks to establish a clear understanding of the bushfire landscape and identify the constraints, opportunities and planning responses required to facilitate future development while achieving an appropriate level of bushfire resilience.

The intended outcomes of this assessment are to:

- identify the broad bushfire hazard characteristics affecting the precinct, including vegetation, topography, bush fire prone land mapping, fire history and potential fire run influences;
- establish the strategic line of management between future development areas and unmanaged bushfire hazard;
- determine the feasibility of accommodating future Asset Protection Zones (APZs) within the precinct and identify areas where APZ provision may constrain development outcomes;
- inform the allocation of future land uses having regard to bushfire exposure, vulnerability, operational requirements and consequence of failure;
- identify strategic access, egress and emergency management requirements necessary to support evacuation, firefighting operations and precinct resilience;
- guide the location of critical infrastructure and higher vulnerability land uses towards areas of comparatively lower bushfire exposure;
- inform the integration of ecological rehabilitation, vegetation retention and environmental outcomes with bushfire protection requirements;
- establish bushfire-related staging requirements to ensure that future development can be delivered in a safe, logical and compliant manner; and
- provide a framework for ongoing engagement with the NSW Rural Fire Service and other agencies throughout the planning process.

Importantly, this assessment recognises that bushfire resilience extends beyond the provision of APZs. At a strategic planning level, resilience is achieved through appropriate land use planning, the establishment of defendable development interfaces, the integration of access and emergency response infrastructure, the management of future vegetation interfaces, and the creation of a precinct structure capable of adapting to future bushfire risk, including the increasing influence of climate change. This includes the establishment of a connected road network linking future structure plan areas, providing multiple access opportunities and improving network resilience during emergency events.

Consistent with the principles of PBP, the road network has been designed where practicable to incorporate perimeter roads along the bushland interface to assist in separating future development from unmanaged vegetation, facilitate firefighting operations, support APZ implementation and maintenance, and provide alternative access and egress opportunities for both emergency services and occupants during bushfire events. The objective is to create a precinct structure that reduces reliance on single points of access, supports evacuation and emergency response, and contributes to the long-term resilience and safety of future development.

The assessment therefore translates the requirements and objectives of PBP into strategic planning considerations, master plan objectives and future development controls. In doing so, it provides a framework to ensure that bushfire risk is appropriately considered in future zoning, subdivision, infrastructure planning and land use decisions, while avoiding unnecessary duplication of development application stage assessment and design processes.

2.3 Required outcomes

The bushfire assessment is intended to provide an evidence-based foundation for the rezoning and Master Plan process by confirming the bushfire hazard context across the precinct, including vegetation, topography, hazard continuity and the areas where bushfire risk may materially influence land use intensity, development yield and siting outcomes. The assessment will undertake a strategic review of Asset Protection Zone (APZ) feasibility and associated land take requirements to assist in verifying developable land and informing the intended character and function of individual sub-precincts. It will also identify strategic evacuation, emergency access and egress considerations, including the interaction between access constraints, development staging and bushfire conditions, and will provide preliminary advice regarding firefighting water supply requirements as an input to infrastructure planning, servicing strategies and staging considerations.

3. Bushfire Resilience Objectives

The Draft Structure Plan adopts a bushfire resilience framework that is founded on a constraints-first approach to planning. Under this approach, bushfire risk is recognised as a primary land use planning consideration that influences the location, design and staging of future development, rather than a constraint to be resolved through development-stage mitigation measures. The objective is to create a precinct that is capable of avoiding, accommodating and responding to bushfire risk through its fundamental structure, land use pattern and infrastructure network.

The bushfire resilience objectives for the Master Plan are to:

- Locate future development in areas capable of achieving an appropriate level of bushfire protection having regard to vegetation, topography, fire run potential and future rehabilitation outcomes.
- Establish a clear and enduring line of management between development areas and unmanaged bushfire hazard, creating a defendable and maintainable development interface.
- Ensure sufficient land is available to accommodate Asset Protection Zones (APZs), perimeter roads, emergency access corridors and other bushfire protection measures without compromising the strategic intent of the precinct.
- Minimise the extent and complexity of development interfaces with bushland and future rehabilitation areas through appropriate land use allocation and precinct design.
- Integrate APZs, managed land, open space, infrastructure corridors and perimeter roads into the overall structure of the precinct to provide separation from bushfire hazard and improve operational resilience.
- Provide a connected and resilient road network that supports firefighting operations, emergency response, evacuation and alternative access opportunities during bushfire events.
- Facilitate the provision of reliable firefighting water supplies and supporting infrastructure as an integral component of future servicing and staging strategies.
- Locate higher vulnerability and higher consequence land uses within areas of comparatively lower bushfire exposure and provide an appropriate graduated risk profile across the precinct.
- Ensure rehabilitation, biodiversity and environmental outcomes are planned in conjunction with bushfire protection requirements so that future vegetation management outcomes do not compromise the safety or functionality of the precinct.
- Deliver development in a staged manner that ensures bushfire protection measures, including APZs, access arrangements, emergency infrastructure and water supplies, are established prior to occupation of each stage.
- Support a range of emergency management responses, including firefighting operations, operational continuity, defend-in-place opportunities where appropriate, and evacuation during larger bushfire events.
- Promote long-term resilience by recognising the potential influence of climate change on future bushfire behaviour and incorporating flexibility, redundancy and adaptive capacity into the structure of the precinct.

Collectively, these objectives seek to ensure that bushfire resilience is embedded within the Master Plan and that future development can proceed within a framework that protects life, supports emergency response, maintains operational functionality and provides a robust basis for future planning and development assessment.

These can be summarised into three core objectives:

Asset Protection Zones and Development Interface

- Future development areas are capable of accommodating appropriate APZs, managed land and defendable space, establishing a clear and maintainable interface between development, rehabilitation areas and bushfire hazard consistent with the principles of PBP.

Or

Land Use Planning and Risk-Based Siting

- Locate future development, critical infrastructure and higher consequence land uses within areas capable of achieving an appropriate level of bushfire protection, having regard to vegetation, topography, fire run potential, access opportunities, rehabilitation outcomes and long-term bushfire risk.

Access and Network Resilience

- Establish a connected and resilient road network that supports firefighting operations, emergency response, maintenance of bushfire protection measures and alternative access opportunities, while reducing reliance on single points of access and improving precinct-wide connectivity.

Emergency Management and Community Resilience

- Provide a precinct structure that supports effective emergency management outcomes through the integration of access, firefighting water supplies, defendable space, emergency response infrastructure and operational flexibility, including both evacuation and defend-in-place options where appropriate.

4. Bushfire Resilience Performance Criteria

The following aim has been developed having regard to the overarching objective of PBP and recognises that bushfire risk is a fundamental consideration in the planning and development of the precinct. Consistent with the constraints-led approach adopted through the EbD process, the aim seeks to ensure that bushfire protection outcomes are balanced with development opportunities, operational requirements, rehabilitation objectives and environmental considerations, establishing a resilient framework to guide future land use planning, infrastructure provision and precinct development.

The aim is:

> *To provide for the protection of human life and minimise impacts on property from the threat of bushfire, while having due regard to development potential, site characteristics, operational requirements, rehabilitation outcomes and the protection of environmental values.*
>
> *To enable ongoing land management practices that support bushfire resilience, ecological sustainability and the long-term health and diversity of rehabilitated and retained landscapes.*

The following performance criteria establish the bushfire resilience framework for the Draft Structure Plan and provide the basis for integrating bushfire considerations into future land use planning, infrastructure provision, development staging and precinct design.

The criteria have been developed having regard to PBP, the strategic planning principles of *Ministerial Direction 4.3 – Planning for Bushfire Protection*, and the findings of the bushfire assessment undertaken for the precinct.

Consistent with the constraints-led approach adopted through the EbD process, the performance criteria seek to ensure that bushfire risk is addressed through the structure and layout of the precinct, the provision of APZs, access and emergency management infrastructure, and the long-term management of the environment and development interfaces.

Collectively, the criteria provide a framework for achieving a resilient, functional and developable precinct while balancing environmental, operational and community outcomes.

4.1 Land Use Planning and Risk-Based Siting

Development should appropriately respond to bushfire risk through:

i. locating future development within areas capable of achieving an appropriate level of bushfire protection having regard to vegetation, topography, fire run potential, rehabilitation outcomes and access opportunities;
ii. locating higher vulnerability and higher consequence land uses within areas of comparatively lower bushfire exposure;
iii. avoiding development patterns that create isolated pockets of development or increase reliance on a single access route;
iv. minimising the extent of development interfaces exposed to unmanaged bushfire hazard;
v. minimising vegetated corridors that facilitate bushfire spread towards and within development areas;
vi. considering rehabilitation and vegetation management outcomes as part of long-term bushfire risk management.

4.2 Asset Protection Zones and Development Interfaces

Development should provide for bushfire protection through:

i. establishing a clear and enduring line of management between development and bushfire hazard;
ii. providing sufficient land to accommodate future APZs;

iii. integrating APZs into the structure of the precinct through managed land, open space, road corridors, utility corridors and other compatible land uses;
iv. ensuring APZs can be practically established, managed and maintained over the life of the development;
v. minimising land use conflicts between APZ requirements, environmental outcomes and rehabilitation objectives.

4.3 Access, Egress and Network Resilience

Development should provide a connected and resilient access network through:

i. establishing a road hierarchy that provides connectivity between precincts and supports emergency response operations;
ii. reducing reliance on single points of access wherever practicable;
iii. providing alternative access opportunities capable of supporting emergency services and evacuation movements;
iv. incorporating perimeter roads along development interfaces where practicable to support APZ management, emergency access and firefighting operations;
v. ensuring road infrastructure can accommodate the increased demands placed upon the network during emergency events;
vi. integrating access planning with development staging to ensure emergency access arrangements are available prior to occupation.

4.4 Emergency Management and Community Resilience

Development should support effective emergency management outcomes through:

i. providing a precinct structure that supports evacuation, defend-in-place and operational continuity options where appropriate;
ii. ensuring development precincts have sufficient separation from bushfire hazard to support emergency response operations;
iii. facilitating coordinated emergency management planning across the precinct;
iv. considering the interaction between future development stages, population growth and emergency management requirements;
v. ensuring emergency response arrangements can adapt to a range of bushfire scenarios, from localised incidents through to major landscape-scale bushfires.

4.5 Firefighting Water Supply and Infrastructure Resilience

Development should provide for the protection of life, property and critical infrastructure through:

i. integrating firefighting water infrastructure with the road network, development interfaces and perimeter access routes;
ii. ensuring firefighting water supplies are available as development stages are delivered;
iii. providing resilient infrastructure networks that maintain functionality during bushfire events;
iv. considering redundancy and alternative supply arrangements for critical infrastructure where appropriate.

4.6 Rehabilitation, Environment and Bushfire Integration

Development should integrate bushfire protection and environmental outcomes through:

i. ensuring rehabilitation and vegetation retention strategies are compatible with bushfire protection objectives;
ii. recognising that future vegetation growth and fuel accumulation may influence long-term bushfire risk;
iii. avoiding the creation of unmanaged vegetation interfaces that increase exposure to future development;
iv. integrating environmental management areas, open space and bushfire protection measures into a coordinated landscape framework;
v. establishing management arrangements that support both ecological and bushfire resilience outcomes.

4.7 Landscape Management and Ecological Resilience

Development should support the long-term management of bushfire risk and environmental values through:

i. establishing a landscape management framework that integrates bushfire protection, rehabilitation and biodiversity outcomes;
ii. recognising the role of fire as a natural ecological process within appropriate vegetation communities and landscape settings;
iii. applying a risk-based approach to fuel management that prioritises the protection of life, property and critical infrastructure while maintaining environmental values;
iv. promoting a diversity of vegetation age classes, structure and habitat values where compatible with bushfire protection objectives;
v. ensuring rehabilitation and vegetation management strategies do not create unacceptable levels of bushfire risk to future development;
vi. facilitating ongoing fuel management, ecological burning and other land management practices where appropriate to support both bushfire resilience and ecological sustainability;
vii. maintaining long-term vegetation management arrangements that support landscape health, biodiversity conservation and the operational requirements of the precinct.

4.8 Climate Adaptation and Long-Term Resilience

Development should support long-term bushfire resilience through:

i. recognising the potential influence of climate change on future bushfire behaviour, fire weather and emergency management requirements;
ii. incorporating flexibility into land use planning, infrastructure provision and staging to respond to changing risk profiles;
iii. avoiding development outcomes that create future constraints on bushfire management;
iv. promoting a precinct structure capable of adapting to future environmental, operational and emergency management challenges.

It is expected that these performance criteria will be reviewed as part of the documentation process to align with other functional areas as required.

5. Evidence base for rezoning and Master Plan

The bushfire assessment seeks to establish a strategic understanding of the bushfire hazard context affecting the precinct, including the influence of vegetation, topography, fuel continuity and broader landscape factors on future development.

The assessment has been undertaken to identify areas where bushfire may influence land use allocation, development intensity and siting, verify that sufficient land is available to accommodate future APZs and associated bushfire protection measures, and inform the development of a resilient precinct structure.

The assessment also considers strategic access, evacuation and emergency response requirements, including the interaction between access networks, development staging and bushfire conditions, together with the provision of firefighting water supply infrastructure as a key servicing and staging consideration. Collectively, these outcomes provide the evidence base necessary to support developable land verification, inform sub-precinct planning and establish a framework for integrating bushfire resilience into the Draft Structure Plan.

5.1 Bush Fire Prone Land

The site is identified as Bush Fire Prone Land (BFPL) for the purposes of Section 10.3 of the *Environmental Planning and Assessment Act 1979* (EPA Act), and the legislative provisions applicable to development on bush fire prone land apply to future development within the precinct (refer Figure 2).

Bush Fire Prone Land mapping provides the statutory trigger for the consideration of bushfire risk during future planning and development assessment processes. Bush Fire Prone Land is land identified by the Council as being capable of supporting bushfire or being subject to bushfire attack. These maps are prepared by the local Council and certified by the Commissioner of the NSW Rural Fire Service (RFS).

The current Bush Fire Prone Land mapping indicates that large portions of the precinct are affected by Vegetation Category 1 bush fire prone land, with smaller areas affected by Vegetation Category 3 vegetation and associated vegetation buffers. The extent of mapped bush fire prone land reflects the presence of both remnant vegetation and areas capable of supporting future bushfire behaviour within and surrounding the precinct.

It is acknowledged that portions of the site comprise active and historical mining areas where vegetation has been removed or substantially modified and, from a bushfire hazard perspective, may not exhibit the characteristics typically associated with bush fire prone land. However, for the purposes of strategic planning and master plan development, the mapping provides an appropriate and conservative basis for assessing bushfire risk across the broader precinct.

At the master planning level, the significance of the mapping extends beyond its statutory function as a development trigger. The mapping assists in identifying areas where bushfire is likely to influence future land use allocation, rehabilitation outcomes, development interfaces, APZ requirements, access and evacuation planning, infrastructure corridors and staging. Accordingly, the mapping has been used as an important strategic planning tool to ensure that bushfire considerations are embedded within the development of the Draft Structure Plan and the long-term planning framework for the precinct.

5.2 Asset Protection Zones

At the strategic planning level, APZs represent one of the primary determinants of developable land capacity and therefore have been considered as part of the Master Plan with detail to be provided in future development applications. PBP requires development to achieve appropriate separation from bushfire hazard through the provision of APZs, with the required extent of separation varying depending on vegetation type, slope, land use and development vulnerability.

A high-level assessment of APZ feasibility has been undertaken to inform the development of the Draft Structure Plan and verify that sufficient land is available to accommodate future bushfire protection requirements. This assessment has not sought to determine detailed APZ dimensions or development application stage compliance outcomes. Rather, it has considered whether the proposed structure planning framework can reasonably accommodate future APZ requirements and maintain flexibility for detailed design and land use planning.

The precinct benefits from its substantial landholding, which provides opportunity to establish appropriate separation between future development areas and bushfire hazard. The Draft Structure Plan has therefore been prepared having regard to the likely requirement for APZs around future development precincts, allowing for the integration of defendable space, perimeter road opportunities, access corridors, open space, natural barriers such as water bodies and vegetation management interfaces within the broader land use framework.

At this stage, the assessment indicates that the identified sub-precincts are generally capable of accommodating future APZ requirements consistent with the principles and intent of PBP. The scale of the site provides flexibility to locate development away from higher hazard areas, establish manageable development interfaces and incorporate bushfire protection measures without materially constraining the strategic intent of the Master Plan.

Importantly, APZs should not be viewed solely as a land constraint. At a precinct planning scale, APZs also provide opportunities to establish integrated landscape outcomes, perimeter access routes, emergency management infrastructure, ecological transition zones and strategic separation between development and future rehabilitation areas. APZs have been incorporated into the broader structure planning framework and utilised to support both bushfire resilience and place-making outcomes.

Consistent with PBP, the Master Plan incorporates the provision of managed land surrounding future development areas to provide separation from bushfire hazard and facilitate the establishment of APZs. Managed land may include a range of compatible uses and landscape elements, including roads, landscaped open space, recreation areas, playing fields, drainage corridors, utility corridors, parklands, environmental management areas subject to appropriate fuel management, and other regularly maintained lands. These areas can perform a dual function by contributing to the amenity and operation of the precinct while also providing defendable space and reducing bushfire exposure. The use of managed land as part of the APZ framework enables future development areas to be buffered from unmanaged vegetation and rehabilitation areas, supports access for maintenance and firefighting operations, and assists in creating a clear and enduring line of management between development and bushfire hazard.

Detailed APZ requirements, including final widths, management regimes and any performance-based justification, will require further refinement as land uses, development footprints, rehabilitation outcomes and vegetation management strategies are confirmed through subsequent planning and development assessment stages. However, the strategic assessment demonstrates that the precinct contains sufficient land to accommodate future APZ requirements and that bushfire protection measures will not represent a fundamental constraint to the implementation of the Draft Structure Plan.

5.3 Evacuation and emergency access

Access and egress represent critical components of bushfire resilience and have been considered as a primary design requirement throughout the EbD process and development of the Draft Structure Plan. PBP recognises that the road network performs multiple functions during a bushfire event, including facilitating evacuation, supporting firefighting operations, providing access to defendable space and APZs, and enabling emergency services to safely access affected areas. As such, access planning has been addressed at the strategic planning stage and integrated into the overall structure and layout of the precinct.

Through the EbD process, access and movement networks have been considered alongside land use planning, environmental constraints, rehabilitation outcomes and infrastructure requirements to establish a connected and resilient road hierarchy across the precinct. This collaborative approach has enabled the testing of alternative access arrangements and the identification of strategic linkages between existing and future development areas, ensuring that access considerations have informed the evolution of the Structure Plan.

Consistent with the strategic planning principles of PBP, the Draft Structure Plan seeks to provide a connected road network that links future sub-precincts and reduces reliance on single points of access wherever practicable. The provision of multiple road connections improves network resilience by providing alternative routes for emergency services and occupants during bushfire events, reducing the potential for isolation should a section of the network become compromised by fire, smoke, debris or other emergency conditions.

The Structure Plan also seeks to incorporate perimeter roads and edge-road interfaces where practicable. These roads provide a number of strategic bushfire planning benefits, including establishing a defined development edge, supporting the provision and maintenance of APZs, improving access for firefighting operations, reducing direct development interfaces with unmanaged vegetation and contributing to evacuation and emergency response capability. Perimeter roads can also function as strategic fuel management corridors and assist in establishing a clear and enduring line of management between future development and bushfire hazard.

At the master planning scale, the assessment has focused on the strategic capacity of the road network to support future development rather than detailed road design. Matters such as carriageway widths, turning movements, gradients, passing opportunities, intersection design and emergency vehicle access will require further assessment during subsequent planning and design stages. Nevertheless, the Draft Structure Plan demonstrates that a framework exists to provide access arrangements consistent with the objectives and intent of PBP.

It is recognised that not all bushfire events will necessitate evacuation of the precinct. Many bushfires, particularly smaller incidents occurring under moderate weather conditions, can be successfully contained and suppressed by emergency services before they pose a significant threat to occupants, infrastructure or operations. However, larger bushfires occurring under elevated or extreme fire weather conditions may result in a heightened risk environment where emergency management actions, including partial or full evacuation, movement restrictions, shelter-in-place arrangements or operational shutdown procedures, may be required. Accordingly, the access network has been considered not only from an evacuation perspective but also from the broader perspective of emergency response, operational continuity and community resilience.

The scale of the precinct and the spatial separation achieved through the Draft Structure Plan provide additional resilience benefits. The proposed development precincts generally incorporate substantial depth, separation from unmanaged vegetation and opportunities for defendable space through the provision of APZs, managed land and road corridors. These characteristics create opportunities for defend-in-place strategies, temporary refuge, operational continuity and staged emergency response measures where appropriate, particularly for industrial and infrastructure land uses with trained personnel and established emergency management procedures. While detailed emergency management arrangements will be developed during future planning and operational stages, the Structure Plan establishes a framework that supports a range of emergency response options and reduces reliance on evacuation as the sole bushfire risk management measure.

Development staging has also been considered as part of the bushfire assessment. The timing and sequencing of future development has the potential to influence access, evacuation and emergency response outcomes if critical infrastructure is delivered after occupation occurs. Accordingly, future stages should ensure that access roads, APZs, emergency management infrastructure and firefighting water supplies are delivered in a coordinated manner and are operational prior to occupation of each stage. No stage should rely on future development to achieve acceptable levels of bushfire protection or emergency access.

The strategic objective is to create a precinct-wide access network that remains functional under emergency conditions, supports firefighting and evacuation operations, reduces the potential for isolated development outcomes and contributes to the long-term resilience of the precinct. Through the EbD process, the Draft Structure Plan has established the framework necessary to support these outcomes and provide a robust basis for future detailed planning and design.

5.4 Firefighting water supply

The provision of reliable firefighting water supply is a fundamental component of bushfire resilience and has been considered as part of the strategic planning framework for the precinct. PBP requires development to be supported by adequate water supplies capable of assisting firefighting operations and contributing to the protection of life, property and critical infrastructure during bushfire events. At the master planning scale, the primary consideration is ensuring that the future structure of the precinct can accommodate a water supply network capable of meeting these requirements.

The Draft Structure Plan provides a framework for the integration of water supply infrastructure with future road networks, development precincts and emergency access arrangements, ensuring that firefighting water can be delivered in a coordinated and efficient manner as development progresses.

At this stage, the assessment has focused on the strategic feasibility of servicing future development areas rather than detailed hydraulic design. The Draft Structure Plan identifies sufficient opportunity for the provision of reticulated water infrastructure, dedicated firefighting water storage, static water supplies, hydrant systems, water supply corridors and other emergency water infrastructure that may be required to support future development. The final form of water supply infrastructure will be determined through subsequent planning, servicing and engineering investigations having regard to the nature of the proposed land uses, development staging and operational requirements.

For a precinct of this scale, firefighting water supply has been considered as a critical component of broader infrastructure resilience. Future planning will consider the need for redundancy, network reliability and emergency operation under bushfire conditions, including the potential provision of multiple water sources, protected infrastructure and strategically located water access points capable of supporting firefighting operations during emergency events.

Development staging will also play an important role in the delivery of firefighting water infrastructure. Consistent with the principles of PBP, each stage of development should be supported by an operational firefighting water supply at the time of occupation or commencement of use. Staging should not rely on future infrastructure upgrades to achieve bushfire protection outcomes. Rather, firefighting water supplies should be delivered in parallel with development, access infrastructure and APZs to ensure that each stage is capable of functioning safely and independently under bushfire conditions.

Where perimeter roads and development interface areas are provided as part of the bushfire protection framework, these areas should be supported by accessible firefighting water supplies consistent with the requirements of PBP and the relevant Australian Standards, including *AS 2419.1 – Fire Hydrant Installations*. Firefighting water infrastructure should be strategically located to support emergency response operations within developed areas along the bushland interface, provide access for firefighting appliances operating from perimeter roads, and facilitate the protection of development precincts during bushfire events. The detailed location, capacity and design of firefighting water infrastructure will be determined during future servicing and engineering investigations.

The strategic assessment indicates that the Draft Structure Plan provides sufficient flexibility to accommodate future firefighting water supply requirements and that water servicing is unlikely to represent a constraint to the implementation of the Structure Plan. Instead, firefighting water supply should be viewed as a key servicing input that will inform future infrastructure planning, precinct staging and the detailed design of development within the precinct.

6. Potential Bush Fire Prone Land Activation Framework

As part of the strategic planning process, consideration should be given to the development of a planning mechanism that recognises the distinction between land that is identified as bush fire prone and land where bushfire exposure has been appropriately addressed through the implementation of permanent bushfire protection measures secured as part of the Structure Plan. The concept to be tested is whether areas of the precinct that have been "activated" for development and are protected through integrated bushfire mitigation measures may warrant a different planning treatment to undeveloped or unmanaged bush fire prone land.

The basis of this approach is that Bush Fire Prone Land mapping identifies the presence of bushfire hazard, whereas strategic planning seeks to manage exposure to that hazard. Bushfire risk is not determined solely by the presence of vegetation within the broader landscape, but by the relationship between future development and that vegetation. Where development precincts are separated from retained, remnant or rehabilitated bushland through the provision of permanent and enduring bushfire protection measures, the practical exposure of those lands to bushfire attack may be substantially reduced. Such measures may include perimeter roads, APZs, managed land, open space corridors, infrastructure corridors, water bodies, strategic fuel management areas, emergency access routes and other landscape elements that establish a clear and maintainable line of management between development and bushfire hazard.

Under this concept, consideration could be given to a planning framework whereby the Bush Fire Prone Land designation remains associated with retained bushland, rehabilitation areas and vegetation corridors that continue to present a bushfire hazard, while land that has been activated and protected through approved and secured bushfire mitigation measures is recognised as having appropriately addressed the strategic and development assessment bushfire planning requirements. The intent would not be to remove the need for bushfire assessment where future development remains exposed to hazard, but rather to acknowledge where the Structure Plan has already secured the necessary separation, access, emergency management and bushfire protection measures at a precinct scale.

Such an approach offers a number of potential strategic planning benefits. It reinforces the principle that bushfire is fundamentally a land use planning issue that is most effectively addressed through the structure and layout of development rather than repeatedly reassessed through subsequent planning proposals where the relevant protection measures have already been established. It would provide greater certainty regarding development capacity, infrastructure planning, investment decisions and precinct activation, while recognising the substantial investment required to deliver perimeter roads, APZs, managed interfaces, firefighting infrastructure and other long-term bushfire protection measures.

The concept also aligns with the broader objectives of the PMLU program and the NSW Government's commitment to facilitating the timely activation and transition of former mining lands to productive future uses. A key objective of the master planning process is to remove unnecessary barriers to development where risks have been appropriately identified, planned for and managed through the strategic planning framework. By recognising bushfire protection measures that have been secured through the Structure Plan and implemented as part of precinct activation, there is potential to streamline future planning and assessment pathways while maintaining appropriate levels of community safety and bushfire resilience.

Importantly, this approach is not without precedent. A similar outcome is achieved through the NSW Rural Fire Service process for the review and amendment of Bush Fire Prone Land mapping within urban release areas and other developing urban environments. In these circumstances, Bush Fire Prone Land mapping may be amended where development has established permanent and maintainable separation from bushfire hazard through urban infrastructure, roads, managed land, APZs and other approved planning measures. The principle underpinning both approaches is that bushfire risk is most effectively managed through strategic land use planning and the creation of enduring development interfaces, rather than through the continued application of bush fire prone land mapping to areas where exposure to bushfire hazard has already been mitigated.

Accordingly, the proposed activation framework seeks to apply a similar risk-based and outcome-focused planning philosophy to large-scale post-mining land use projects. The framework recognises the transition of land from a bushfire-prone landscape to a managed and protected urban, employment or mixed-use precinct where the necessary bushfire protection measures have been secured and are capable of being maintained in perpetuity. Any future implementation mechanism would require further investigation and consultation with the NSW Rural Fire Service, Department of Planning, Housing and Infrastructure and relevant consent authorities to ensure that community safety outcomes are maintained, bushfire protection measures remain secured and enforceable over time, and future changes in vegetation, rehabilitation outcomes or land management practices do not compromise the level of protection achieved.

At this stage, the concept is presented as a strategic planning initiative for further testing and discussion. The objective is to determine whether a more proportionate, risk-based and resilience-focused approach to Bush Fire Prone Land mapping and planning controls can be developed for activated precincts where bushfire protection outcomes have already been embedded within the Structure Plan and secured prior to development occurring.

7. Conclusion

Bushfire has been identified as a primary design driver for the future planning and activation of the Macquarie Mine precinct. The site is located within a broader landscape capable of supporting bushfire activity and contains areas mapped as Bush Fire Prone Land. As a result, bushfire risk has been considered as a fundamental input into the development of the Draft Structure Plan rather than a matter to be addressed through subsequent development assessment processes.

The assessment has been undertaken using a constraints-led and risk-based approach consistent with the objectives of Planning for Bush Fire Protection 2019 (PBP), the Enquiry by Design (EbD) process and the broader strategic planning framework established for the Post Mining Land Use program. Bushfire considerations have been integrated alongside rehabilitation planning, biodiversity outcomes, infrastructure requirements, access networks, servicing opportunities and future land use objectives to ensure that bushfire resilience is embedded within the structure and function of the precinct.

The strategic assessment confirms that the Draft Structure Plan is generally capable of accommodating future bushfire protection measures, including Asset Protection Zones (APZs), managed land, perimeter road opportunities, emergency access arrangements and firefighting infrastructure. While detailed APZ design has not been undertaken due to the strategic nature of the assessment and the absence of defined development footprints, sufficient land appears available to accommodate future APZ requirements consistent with the principles and intent of PBP. Bushfire protection requirements are therefore not expected to represent a fundamental constraint to future land use allocation or precinct activation.

The assessments undertaken the site provide a practical test of the bushfire planning principles embedded within the Draft Structure Plan. The assessments indicate that future development can generally be configured to respond appropriately to the surrounding bushfire hazard through the provision of suitable separation from hazard, managed land, APZs, integrated access and emergency management infrastructure, firefighting water supply and ongoing vegetation management. The Structure Plan provides sufficient flexibility to accommodate future bushfire protection measures while supporting development activation, rehabilitation objectives and long-term land use outcomes. While detailed APZ design, Bushfire Attack Level (BAL) assessments and development-specific compliance measures will be required during future planning and development stages, the strategic assessment has not identified any fatal bushfire constraints affecting the area. The findings provide confidence that the bushfire resilience framework established through the Draft Structure Plan is capable of supporting future precinct activation and development while maintaining appropriate levels of community safety and bushfire resilience.

The assessment has demonstrated that access, egress and emergency management considerations can be accommodated within the Draft Structure Plan through the provision of an integrated road hierarchy, multiple access opportunities, managed development interfaces and future emergency management planning. Significant work was completed through the development of the Draft Structure Plan to provide road linkages that meet the intent of Planning for Bushfire Protection. Similarly, firefighting water supply requirements can be incorporated into future servicing and infrastructure strategies and do not represent a strategic limitation to future development.

Importantly, the assessment recognises that bushfire resilience extends beyond the provision of APZs and emergency infrastructure. The rehabilitation of the site presents a significant opportunity to integrate bushfire management, biodiversity conservation and ecological resilience outcomes through a coordinated landscape management framework. The strategic planning process has identified opportunities to utilise fire and fuel management as part of a broader landscape stewardship approach that supports ecological function, rehabilitation objectives and long-term bushfire resilience.

Based on the strategic assessment undertaken, no fatal bushfire constraints have been identified that would prevent the implementation of the Draft Structure Plan. Bushfire risk can be appropriately managed through strategic land use planning, the provision of APZs and managed interfaces, integrated access and emergency management infrastructure, appropriate servicing arrangements and ongoing landscape management practices.

The findings of this assessment support the continued progression of the Draft Structure Plan and provide a strategic framework for incorporating bushfire resilience into future planning and development outcomes. Notwithstanding, further detailed bushfire assessment will be required as future precincts are refined and development proposals progress. This will include detailed consideration of vegetation and slope classifications, APZ design, Bushfire Attack Level (BAL) assessments, access and water supply design, emergency management arrangements and compliance with the relevant version of Planning for Bush Fire Protection and associated statutory requirements applying at the time of development.

Accordingly, subject to future precinct planning, infrastructure delivery, APZ provision, access arrangements, rehabilitation outcomes and detailed development assessment, no fatal bushfire constraints have been identified that would prevent the implementation of the Draft Structure Plan or the long-term activation of the Macquarie Mine precinct.

Figure 1 Structure Plan

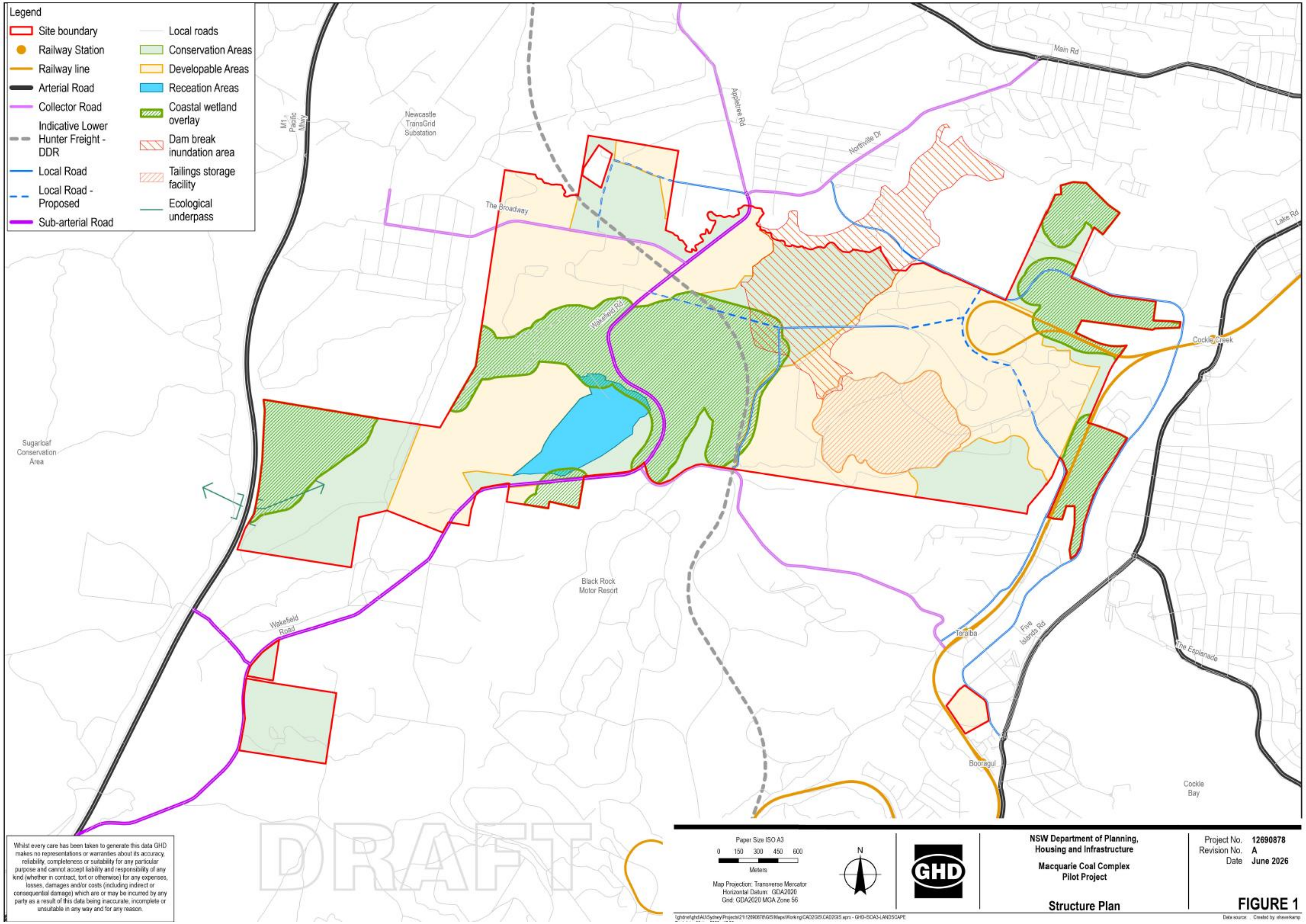
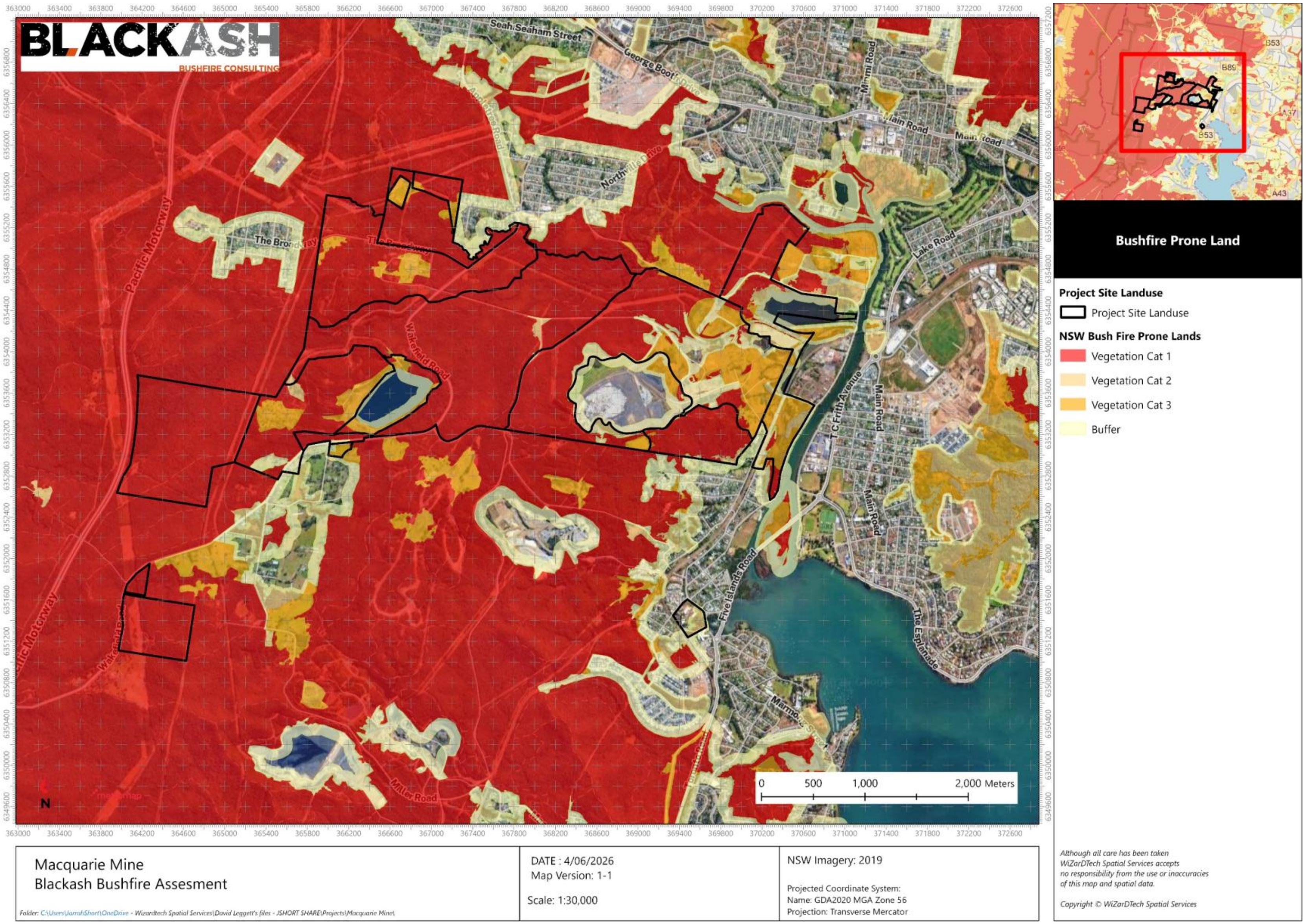


Figure 2 Bushfire Prone Land



Appendix 1 Overview - Strategic Planning Context and Supporting Technical Studies

The preparation of the Draft Structure Plan has been informed by a series of technical investigations and specialist studies that collectively establish the opportunities, constraints and risk profile of the precinct. These studies provide the evidence base for the EbD process and support a constraints-led approach to planning, ensuring that future development is informed by a comprehensive understanding of environmental, operational, infrastructure and hazard considerations.

The appendices to this report provide important background information and technical context that has influenced the development of the Draft Structure Plan. These studies should not be viewed as standalone assessments but rather as interconnected components of a broader planning framework that seeks to balance development opportunities, environmental stewardship, operational requirements, community resilience and long-term land management outcomes.

A key objective of the planning process has been to ensure that future development is located within areas capable of accommodating the necessary infrastructure, environmental protections and risk management measures while maintaining flexibility for future land uses and rehabilitation outcomes. This approach is consistent with the objects of the *Environmental Planning and Assessment Act 1979* (EPA Act), particularly those relating to resilience to natural hazards, ecologically sustainable development, the proper management of natural resources, risk-based decision making and the orderly and economic development of land.

Bushfire has been identified as a primary design driver for the precinct and has therefore been considered at the earliest stages of planning. The bushfire assessment and associated fire history analysis demonstrate that the site is located within an active bushfire landscape that has experienced both wildfire and prescribed burning activity over an extended period. The historical occurrence of fire throughout the surrounding landscape reinforces the need to consider bushfire as a long-term and recurring planning consideration rather than an isolated development constraint. This understanding has informed the location of development precincts, the establishment of development interfaces, access planning, emergency management considerations and the provision of future APZs.

The appendices also recognise the important role that landscape management and ecological resilience play in the long-term success of the precinct. As rehabilitation progresses and vegetation communities mature, the interaction between bushfire risk, biodiversity values, ecological function and land management will become increasingly important. Fire is recognised not only as a hazard that must be managed, but also as a natural ecological process that contributes to the health, diversity and resilience of many Australian ecosystems. Consequently, the planning framework seeks to support a risk-based approach to fuel management that protects life, property and critical infrastructure while also maintaining environmental values, ecological function and biodiversity outcomes.

Importantly, the development of the Draft Structure Plan has not been based on any single constraint or technical discipline. Rather, it reflects a balanced planning response that integrates bushfire risk, rehabilitation requirements, environmental values, infrastructure provision, access and servicing needs, operational considerations and future development opportunities. Through the EbD process, specialist inputs have been brought together to identify areas suitable for development, areas requiring protection or management, and the infrastructure necessary to support a resilient and functional precinct.

The resulting Structure Plan seeks to achieve an appropriate balance between development potential, environmental protection, operational requirements and community resilience. It recognises that long-term success will depend upon the integration of risk management, landscape stewardship and sustainable development principles, ensuring that future land uses can coexist with the environmental and operational characteristics of the site while remaining adaptable to future changes in climate, vegetation, land management practices and community needs.

The following appendices provide the technical foundation for these planning outcomes and should be read as supporting documents that inform the development of the bushfire aspects of the Draft Structure Plan and its associated objectives, performance criteria and implementation framework.

Appendix 2 Bushfire History

The site and surrounding region have an extensive history of bushfire activity, reflecting the broader fire-prone characteristics of the Lower Hunter landscape (Figure 5). Review of available fire history mapping indicates that the site is located within a landscape that has experienced multiple wildfire and prescribed burning events over recent decades, with numerous recorded fire events occurring within 2 km, 5 km and 10 km of the precinct.

The fire history mapping demonstrates that bushfire activity is not isolated to a single portion of the landscape, but occurs across a broad area surrounding the site. Significant wildfire events have historically occurred to the south, south-west and west of the precinct, including large landscape-scale fires extending through the vegetated ridgelines, conservation lands and bushland areas that surround the Lower Hunter Valley. The mapping also identifies numerous smaller wildfire events throughout the surrounding urban-bushland interface, demonstrating the ongoing interaction between development and bushfire hazard across the region.

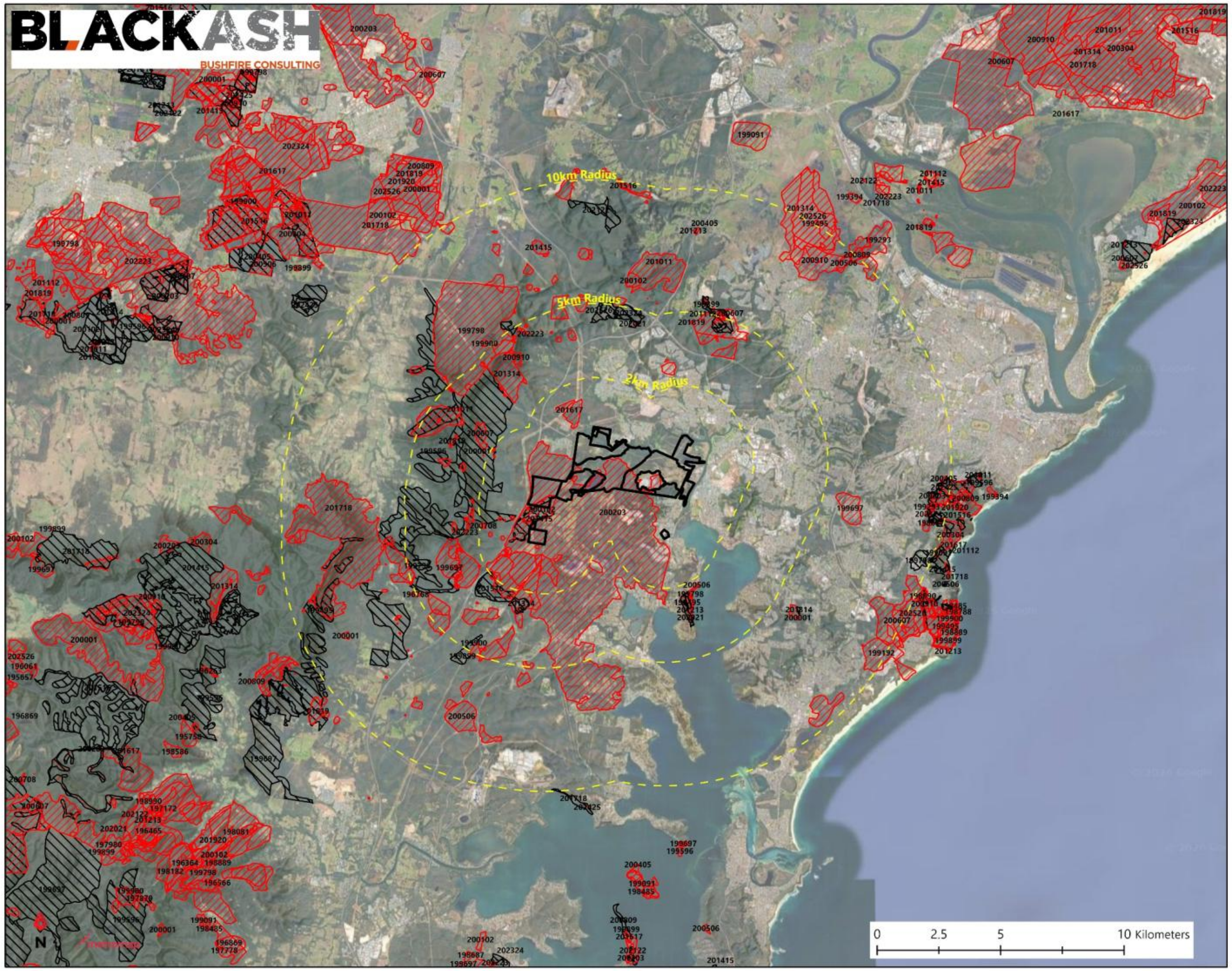
Importantly, the mapping indicates a history of both wildfire and prescribed burning activity within the broader landscape. The occurrence of prescribed burns reflects ongoing fuel management activities undertaken by land managers and emergency service agencies to reduce bushfire risk. While these activities contribute to landscape management and hazard reduction, they do not eliminate the potential for future bushfire events, particularly under severe fire weather conditions.

The pattern of historical fire activity demonstrates that the site is located within an active bushfire landscape capable of supporting both localised and large-scale bushfire events. Historical fires have occurred across a variety of vegetation communities and land tenures, indicating that future bushfire risk is likely to be influenced by broader landscape conditions rather than site-specific factors alone. Consequently, bushfire planning for the precinct has considered the potential for long-duration bushfires, ember attack, smoke impacts and landscape-scale fire runs originating beyond the immediate site boundary.

The Lower Hunter Valley is widely recognised as experiencing elevated bushfire weather conditions, driven by a combination of climatic and landscape factors. The region is subject to frequent periods of high temperatures, low relative humidity and strong westerly to north-westerly winds, particularly during summer and early autumn. These conditions are often exacerbated by prolonged dry periods, resulting in the curing of fine fuels and increased availability of heavier fuels across forested and grassland areas. In a bushfire context, this environment can support rapid rates of fire spread, increased spotting potential and the development of high-intensity fire behaviour.

The documented history of wildfire occurrence surrounding the site reinforces the need for a strategic and precautionary approach to bushfire planning. The Draft Structure Plan has therefore adopted a risk-based framework that recognises bushfire as a primary design driver and seeks to minimise exposure through appropriate land use allocation, the provision of APZs, the establishment of managed development interfaces, the integration of resilient access networks, sustainable approach to integrated fuel management and the incorporation of emergency management infrastructure. The fire history of the surrounding landscape provides strong evidence that future bushfire events should be anticipated as part of the long-term planning and management of the precinct rather than regarded as infrequent or exceptional occurrences.

Figure 3 Bushfire History



Fire History

Project Site Landuse

Project Site Landuse

2km,5km,10km Buffer

2km,5km,10km Buffer

NPWS Fire History

Prescribed Burn

Wildfire

Macquarie Mine Blackash Bushfire Assesment	DATE : 4/06/2026 Map Version: 1-1 Scale: 1:150,000	Projected Coordinate System: Name: GDA2020 MGA Zone 56 Projection: Transverse Mercator

Folder: C:\Users\JarrahShort\OneDrive - Wizardtech Spatial Services\David Leggett's files - JSHORT SHARE\Projects\Macquarie Mine\

Although all care has been taken WiZarDTech Spatial Services accepts no responsibility from the use or inaccuracies of this map and spatial data.

Appendix 3 Landscape Management and Ecological Resilience

The long-term success of the precinct will depend not only on the provision of bushfire protection measures, but also on the sustainable management of the broader landscape within which future development occurs. The site is located within a region where fire has historically been a recurring natural process and where vegetation communities have evolved in response to varying fire regimes over extended periods. As rehabilitation progresses and vegetation communities mature, the interaction between bushfire risk, ecological values and land management practices will become an increasingly important consideration in the long-term management of the precinct.

Bushfire planning is often focused on reducing the likelihood and consequence of fire impacts on life and property. However, from a landscape management perspective, fire also performs an important ecological function. Many Australian vegetation communities rely upon fire as a natural process that contributes to regeneration, habitat diversity, nutrient cycling, succession and ecosystem health. The complete exclusion of fire from the landscape can, in some circumstances, result in declining ecological condition, reduced biodiversity values, altered vegetation structure and the accumulation of fuel loads that may ultimately increase bushfire risk.

Accordingly, the precinct should adopt a landscape management framework that recognises both the hazard and ecological dimensions of fire. This requires an integrated approach where bushfire protection measures are balanced with rehabilitation objectives, biodiversity conservation and long-term environmental stewardship. Rather than seeking to eliminate fire from the landscape, the objective should be to manage fire in a manner that protects life, property and critical infrastructure while maintaining healthy, resilient and functioning ecosystems.

The rehabilitation of former mining lands provides an opportunity to establish a diverse and resilient landscape that incorporates a variety of vegetation communities, habitat values and ecological functions. Where compatible with bushfire protection objectives, consideration should be given to promoting a diversity of vegetation structures, age classes and fuel profiles across the landscape. Such an approach can contribute to ecological resilience while also reducing the likelihood of extensive areas of uniformly aged vegetation capable of supporting large, continuous fire runs.

A risk-based approach to fuel management should be adopted across the precinct. Areas adjoining future development, critical infrastructure and access networks will require a higher degree of management to maintain defendable space, Asset Protection Zones (APZs) and operational access. In contrast, areas located further from development may be managed primarily for ecological outcomes, biodiversity conservation and rehabilitation objectives, subject to appropriate consideration of broader landscape bushfire risk. This graduated management approach enables bushfire protection measures to be targeted where they are most needed while supporting environmental values across the wider landscape.

The use of ecological burning, hazard reduction activities and other landscape management practices should be considered as part of future land management planning. These activities have the potential to contribute to both bushfire resilience and ecological sustainability when implemented in accordance with the characteristics and requirements of the relevant vegetation communities. The objective should be to establish fire regimes that support biodiversity outcomes and ecological function while avoiding the accumulation of fuel loads that may result in unacceptable levels of bushfire risk to development.

Importantly, the landscape management framework should recognise that rehabilitation outcomes will evolve over time. Vegetation communities will mature, fuel loads will change and ecological values will develop throughout the life of the precinct. Ongoing monitoring, adaptive management and periodic review of vegetation management practices will therefore be required to ensure that bushfire protection objectives, ecological resilience and rehabilitation outcomes continue to be achieved over the long term.

Through the integration of bushfire planning, rehabilitation and environmental management, the precinct has the opportunity to establish a landscape that is not only safer and more resilient to bushfire, but also ecologically functional, biodiverse and capable of supporting healthy environmental outcomes for future generations.

Appendix 4 Strategic Planning Context

Strategic planning plays a critical role in achieving the objects of the EPA Act by ensuring that environmental, social, economic and risk considerations are integrated into decision-making before development outcomes are established. At a precinct scale, strategic planning provides the opportunity to identify constraints, opportunities and competing land use demands early in the planning process, allowing future development to be directed towards locations capable of supporting sustainable and resilient outcomes.

The EPA Act recognises that planning should extend beyond the assessment of individual development proposals and instead provide a framework for the orderly and economic use and development of land. This is particularly important for large and complex sites where environmental constraints, infrastructure requirements, natural hazards, rehabilitation objectives and future development opportunities must be considered collectively rather than in isolation.

The site presents a range of opportunities and constraints associated with its historical mining use, rehabilitation requirements, environmental values, bushfire risk, infrastructure provision and future land use aspirations. The preparation of the Draft Structure Plan through the EbD process has enabled these factors to be considered in an integrated manner, ensuring that future development is informed by a comprehensive understanding of site conditions and long-term planning objectives.

Importantly, the EPA Act promotes resilience to climate change and natural disasters through adaptation, mitigation, preparedness and prevention. Within the context of this Master Plan, bushfire represents a key natural hazard that must be addressed through strategic land use planning. The most effective bushfire outcomes are achieved when risk is considered at the earliest stages of planning, allowing development to be located, configured and serviced in a manner that reduces exposure to hazard and supports emergency management outcomes. This approach aligns with the broader objective of creating resilient communities and infrastructure capable of adapting to future environmental and climatic conditions.

The EPA Act also promotes ecologically sustainable development through the integration of economic, environmental and social considerations. Strategic planning provides the mechanism through which these often competing objectives can be balanced. For the precinct, this includes integrating bushfire protection requirements with rehabilitation outcomes, biodiversity conservation, infrastructure provision, economic development opportunities and long-term land management objectives. By considering these matters together, the Master Plan seeks to achieve development outcomes that are both environmentally responsible and economically viable.

The EPA Act further promotes a proportionate and risk-based approach to environmental planning and assessment. The constraints-led methodology adopted for the Master Plan reflects this principle by recognising that not all areas of the site present the same opportunities or level of risk. Through the identification of bushfire hazard, environmental constraints, access requirements and infrastructure considerations, the Draft Structure Plan directs future development towards areas capable of supporting an appropriate level of protection while avoiding unnecessary constraints on land that can be safely and efficiently developed.

Ultimately, strategic planning provides the foundation for achieving the objects of the EPA Act by promoting the proper management, development and conservation of land and resources, supporting resilience to natural hazards, protecting environmental values and facilitating the orderly and economic development of the precinct. The Draft Structure Plan represents the practical application of these principles, establishing a framework that balances development opportunities with environmental stewardship, risk management and long-term community resilience.

Appendix 5 Bush Fire Landscape Assessment

A bush fire landscape assessment considers bushfire risk in the context of the broader surrounding landscape, rather than solely within the boundaries of the site. The assessment examines the potential for bushfire ignition, spread and intensity, together with the interaction of vegetation, topography, prevailing weather conditions, access arrangements and opportunities for fire suppression. This provides an understanding of how bushfires may develop within the surrounding landscape and the implications for future development within the precinct.

The site is located within a broader landscape that contains extensive areas of vegetation capable of supporting bushfire. As demonstrated by the bush fire prone land mapping and historical fire activity within the region, there is potential for landscape-scale bushfires to occur within the surrounding environment and influence the site. The extent and continuity of vegetation surrounding parts of the precinct create the potential for fire runs to develop under adverse fire weather conditions, particularly where large areas of unmanaged vegetation occur in conjunction with prevailing winds and topographic influences.

The severity and intensity of potential fire runs will be influenced by a range of factors including vegetation type, fuel condition, topography, weather and the effectiveness of ongoing landscape management. Areas of continuous vegetation have the capacity to support sustained fire spread and generate elevated radiant heat, ember attack and smoke impacts. However, the broader landscape also contains a range of modified land uses, mining areas, infrastructure corridors, managed lands and fragmented vegetation patterns that influence fuel continuity and can reduce the potential for uninterrupted landscape-scale fire spread in some locations.

The complexity of the terrain, vegetation patterns and rehabilitation areas has also been considered as part of the landscape assessment. As rehabilitation progresses, vegetation communities will continue to evolve, potentially altering fuel characteristics and bushfire behaviour over time. Accordingly, bushfire risk should be considered as a dynamic rather than static condition, requiring ongoing landscape management and adaptive planning responses throughout the life of the precinct.

The ability to access and suppress bushfires is a key consideration in determining overall landscape risk. Consistent with the principles of Planning for Bush Fire Protection 2019 (PBP), the Draft Structure Plan has been developed with an integrated road hierarchy that provides connectivity between precincts, supports emergency response operations and facilitates access and egress during bushfire events. The road network has been planned to provide multiple connections between development areas where practicable, reduce reliance on single access points and support the movement of emergency services throughout the precinct.

Where appropriate, perimeter roads, managed interfaces and Asset Protection Zones (APZs) have also been incorporated into the structure planning framework to assist in defining development edges, supporting firefighting operations and improving opportunities for fire suppression. These measures contribute to reducing landscape-scale bushfire risk by creating defendable areas from which firefighting operations can be undertaken and by limiting the direct interface between future development and unmanaged vegetation.

Overall, the assessment indicates that while the site is located within a broader bushfire-prone landscape capable of supporting significant bushfire events, the strategic planning framework provides opportunities to manage this risk through appropriate land use planning, the establishment of defendable development interfaces, integrated access networks, landscape management and the provision of emergency response infrastructure. These measures support a resilient precinct structure that is capable of responding to both localised bushfire incidents and larger landscape-scale fire events.

Landscape Scale Assessment Tool (LSAT)

Blackash Bushfire Consulting has developed a Landscape Scale Assessment Tool (LSAT) to assist in understanding bushfire risk at the strategic planning and master planning level. The LSAT (Figure 6) is designed to assess bushfire risk within the context of the broader landscape rather than focusing solely on site-specific hazard characteristics. This approach aligns with the strategic planning principles contained within PBP, which recognise that bushfire behaviour, emergency management outcomes and development resilience are influenced by factors extending well beyond individual development sites.

The LSAT considers a range of parameters that influence landscape-scale bushfire risk, including surrounding vegetation, potential bushfire behaviour, fire history, fire run potential, vegetation connectivity, separation from hazard, access to shelter, evacuation opportunities, emergency service access and firefighting water supply arrangements. The tool provides a structured framework for identifying the broader bushfire context within which future development will occur and assists in informing strategic land use planning, infrastructure provision, emergency management planning and development staging.

Application of the LSAT to the precinct indicates that the site is located within a landscape that presents a generally elevated bushfire risk profile. The surrounding landscape contains extensive areas of vegetation capable of supporting bushfire, with the potential for long-distance fire runs originating beyond the site boundary. Historical fire mapping demonstrates a long history of wildfire occurrence throughout the broader Lower Hunter region, reinforcing the conclusion that the site is situated within an active bushfire landscape where both localised and landscape-scale bushfire events should be anticipated.

The assessment identifies the surrounding vegetation and broader bushfire behaviour characteristics as the primary contributors to landscape-scale risk. Large areas of vegetation occur within the broader catchment surrounding the precinct and have the potential to support significant fire development under adverse fire weather conditions. However, the site also benefits from substantial areas of existing disturbance associated with mining operations, infrastructure corridors, managed lands and fragmented fuel patterns which reduce vegetation continuity and provide opportunities to influence future bushfire behaviour through strategic planning and landscape management.

The LSAT assessment also recognises the importance of emergency management infrastructure in moderating landscape-scale risk. The Draft Structure Plan has been developed through an integrated planning process that incorporates a connected road hierarchy, future APZs, managed land, perimeter road opportunities, firefighting water supplies and emergency access arrangements. These measures improve the ability of emergency services to access the site, support firefighting operations and provide alternative access and egress opportunities during bushfire events.

Overall, the LSAT indicates that while the site is located within a broader landscape capable of supporting significant bushfire activity, the risk can be appropriately managed through a combination of strategic land use planning, integrated infrastructure provision, landscape management and emergency management measures. The findings reinforce the importance of the constraints-led approach adopted through the EbD process and support the incorporation of bushfire resilience principles into the long-term planning framework for the precinct.

Figure 4 Landscape Scale Assessment

Landscape Scale Assessment Tool					
Parameter	**Low landscape scale threat**	**Moderate landscape scale threat**	**High landscape scale threat**	**Extreme landscape scape threat**	**Rating**
1. Surrounding Vegetation	Bushfire cannot directly approach the site as it is surrounded by urban development and non-mapped vegetation or managed land.	Bushfire can only approach from one aspect and the site is within a suburban, township or urban area considered managed land. Typically an island of bushfire vegetation within a wider urban development area; site with highly fragmented vegetation nearby; or interface site impacted only by linear vegetation corridors of 100m width or less.	Bushfire can approach from more than one aspect and site is on the bushland-urban interface with the developed area considered as managed land. Typically contiguous bushfire vegetation with a typical fire run in any direction of 0.1-2.0 km distance.	Bushfire can approach from more than one aspect and/or fires have many hours or days to grow and develop before impacting and/or site is surrounded by significant unmanaged vegetation. Typically large areas of contigous bushland with fire runs of more than 2 km possible.	**Extreme**
2. Bushfire Behaviour	Extreme bushfire behaviour at the site is not possible given the broader landscape.	Extreme bushfire behaviour at the site is unlikely in this broader landscape due to combination of factors of vegetation type, vegetation fragmentation, aspect and topography.	Extreme bushfire behaviour at the site is likely in this broader landscape due to combination of factors of vegetation type, vegetation fragmentation, aspect and topography.	Extreme bushfire behaviour is very likely in this broader landscape due to combination of factors of vegetation type, vegetation fragmentation, aspect and topography.	**High**
3. Impact of severe fire behaviour (FFDI 80 or 100 as relevant) coming onto site from wider fire catchment	There is little vegetation beyond 150 metres of the site (except grasslands and low-threat vegetation) and will not result in neighbourhood scale destruction of the site.	The type and extent of vegetation beyond 150m from the site may result in neighbourhood-scale destruction as it interacts with the bushfire hazard on and close to the site.	The type and extent of vegetation beyond 150m is likely to result in neighbourhood-scale destruction as it interacts with the bushfire hazard on and close to the site.	The type and extent of vegetation beyond 150m will result in neighbourhood-scale destruction as it interacts with the bushfire hazard on and close to the site.	**Low**
4. Vegetation Corridors	Vegetation within the site cannot enable fire to enter and move through the site by a continuous fire path from the primary fire source.	Vegetation within the site is unlikely to enable fire to enter and move through the site by a continuous fire path from the primary fire source.	Vegetation within the site may enable fire to enter and move through the site by a continuous fire path from the primary fire source.	Vegetation corridors on site provide for passage of fire to enter and move through the site from the primary fire source.	**Moderate**
5. Separation	Hazard separation between extreme bushfire hazard and buildings of greater than 100m. Extreme bushfire hazard does not include vegetated corridors of less than 100m width or grasslands.	Hazard separation between extreme bushfire hazard and buildings of 50-100m. Extreme bushfire hazard does not include vegetated corridors of less than 100m width or grasslands.	Hazard separation between extreme bushfire hazard and buildings of 20-50m. Extreme bushfire hazard does not include vegetated corridors of less than 100m width or grasslands.	Hazard separation between extreme bushfire hazard and buildings of <20m. Extreme bushfire hazard does not include vegetated corridors of less than 100m width or grasslands.	**Moderate**
6. Shelter	Immediate access is available to a place that provides shelter from bushfire. This includes existing or proposed buildings on site constructed in accordance with PBP.	Access is readily available to a place that provides shelter from bushfire. This will often be the surrounding developed area.	Access to a place that provides shelter from bushfire is not certain during a wildfire and existing buildings are not built to PBP standards.	Access to a place that provides shelter from bushfire is not possible during a wildfire.	**Low**
7. Evacuation	Multiple evacuation routes are available and unlikely to be impacted by fire.	Evacuation to alternate location that provides life safety refuge is <1km and can be completed by foot or vehicle.	Evacuation to alternate location that provides life safety refuge is 1km-10km.	Evacuation to alternate location that provides life safety refuge is > 10km.	**Moderate**
8. Isolation and emergency services	Seamless integration with existing settlement - no impact on evacuation or access for emergency services.	Short bushland pinch points that may carry fire across roads and restrict access briefly during passage of fire. Unlikely impact on evacuation or access for emergency services.	Short bushland pinch points that are likely to carry fire across roads and restrict access temporarily. Likely impact on evacuation or access for emergency services.	Large areas of bushland or multiple pinch points that are likely to carry fire across roads in forest areas and will block evacuation or emergency service access routes for extended time.	**Moderate**
9. Firefighting water supplies	Site is within urban area and has access to reticulated water supply **OR** site has dedicated firefighting water supply in accordance with PBP requirements.	Site is on the periphery of urban area and has access to reticulated water supply that may be more susceptible to interuption.	Site is outside urban area and relies on an on site water supply not in accordance with PBP.	Site is in an isolated area and relies on an on site water supply not in accordance with PBP.	**Moderate**
		Overall Threat Rating	**High Risk**	**Total**	190

Appendix 6 Asset Protection Zones

A strategic APZ feasibility assessment has been undertaken by Blackash as a key component of the bushfire planning process and formed an important consideration throughout the EbD process. Consistent with the constraints-led approach adopted for the preparation of the Draft Structure Plan, the ability to provide and maintain future APZs was considered alongside environmental constraints, rehabilitation objectives, infrastructure requirements, access networks, land use planning and development capacity. This ensured that bushfire protection requirements were integrated into the evolution of the Draft Structure Plan from the outset, rather than being addressed retrospectively during future development assessment stages.

The assessment has considered the broad bushfire hazard context affecting the precinct, including surrounding vegetation, potential development interfaces, future rehabilitation areas, access arrangements and the likely separation distances required to achieve appropriate bushfire protection outcomes. The purpose of the assessment is not to determine detailed APZ dimensions or demonstrate development application stage compliance. Rather, it provides a strategic level verification that sufficient land exists within and surrounding the identified development precincts to accommodate future APZs, managed land, perimeter roads and associated bushfire protection measures consistent with the principles and intent of PBP.

The assessment demonstrates that the scale of the site and the configuration of the Draft Structure Plan provide substantial flexibility to establish defendable space between future development and bushfire hazard. Opportunities exist to integrate APZs within broader landscape and infrastructure outcomes through the provision of managed land, open space corridors, perimeter roads, utility corridors, drainage lands and other compatible land uses. This approach not only facilitates future compliance with bushfire protection requirements but also supports emergency access, landscape management and long-term precinct resilience.

Importantly, APZ feasibility was used as a key land allocation test during the EbD process. Development precincts, infrastructure corridors and future land use interfaces were progressively refined to ensure that bushfire protection requirements could be accommodated without creating unreasonable constraints on development outcomes, rehabilitation objectives or environmental values. Through this process, the Draft Structure Plan has been configured to provide an appropriate balance between development potential, environmental stewardship and bushfire resilience.

Detailed APZs have not been prepared or mapped as part of this assessment. Given the strategic nature of the Master Plan and the absence of defined development footprints, subdivision layouts, building envelopes, final land uses and rehabilitation outcomes, it would be premature to determine detailed APZ widths or management prescriptions at this stage. Such matters are more appropriately addressed during subsequent precinct planning, subdivision design and development application stages when sufficient detail is available to undertake a site-specific assessment in accordance with PBP. The purpose of this assessment is therefore to confirm APZ feasibility and inform strategic land allocation decisions, rather than establish detailed compliance outcomes.

Accordingly, the strategic assessment confirms that future APZ requirements can be accommodated within the Draft Structure Plan and that bushfire protection measures are not expected to represent a fundamental constraint to the implementation of the Master Plan. Detailed APZ design, management requirements and development-specific bushfire protection measures will be refined during subsequent planning, subdivision and development application stages as land uses, building footprints, infrastructure requirements and rehabilitation outcomes become more defined.

Appendix 7 Review of Planning for Bush Fire Protection

The NSW Rural Fire Service (RFS) has advised that Planning for Bush Fire Protection 2019 (PBP) is currently undergoing formal review, with a draft updated document anticipated during 2026 and implementation expected following completion of the review process. While the final content and timing of any revised framework remains subject to government consideration, the review reflects the ongoing evolution of bushfire planning policy in response to changing community expectations, major bushfire events, advances in bushfire science, climate change, infrastructure resilience and contemporary approaches to risk-based land use planning.

The review is particularly relevant to strategic planning projects of this scale and complexity, where development outcomes are expected to endure for decades and where future land use decisions should be capable of responding to changing policy settings and emerging bushfire risk considerations. As a result, the Draft Structure Plan has not been developed solely to satisfy the minimum requirements of the current planning framework. Rather, it adopts a resilience-based and risk-informed approach that seeks to establish enduring bushfire protection outcomes through appropriate land use planning, development interfaces, access arrangements, emergency management infrastructure, landscape management and long-term precinct design.

Importantly, strategic planning provides the opportunity to address bushfire risk before development footprints, infrastructure layouts and land use patterns become fixed. This enables consideration of broader resilience outcomes that extend beyond development application stage compliance, including the integration of APZs, perimeter roads, emergency access networks, firefighting water supplies, rehabilitation outcomes and landscape management frameworks. These measures are inherently adaptable and are likely to remain relevant regardless of future changes to detailed policy requirements.

Given the ongoing review of PBP and the significance of the precinct, consultation with the NSW Rural Fire Service should continue throughout the master planning process to ensure emerging policy directions, strategic planning requirements and agency expectations are appropriately considered. This will assist in developing a Structure Plan that remains robust, adaptable and capable of responding to both current and future bushfire planning requirements.